

Time Allowed: 01:30hrs.

Read the following instructions carefully before attempting the question paper.

1. Match the Paper Code shaded on the OMR Sheet with the Paper code mentioned on the question paper and ensure that both are the same.
2. This question paper contains 60 questions of 1 mark each. 0.25 marks will be deducted for each wrong answer.
3. All questions are compulsory.
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Q1) Analyze the following coding:

In a code language PROPER is coded as NTMRCT, how is RETURN coded in the same code?

- (a) TCVSTL (b) PGRWPP (c) CNBPXU (d) PGRWQB CO6, L4

Q2) Analyze the following coding and answer the question

If rit puc bec means eat fresh food,

puc tec jac means food is tasty,

jac lac mac means she is beautiful,

then which word means tasty?

- (a) puc (b) bec (c) rit (d) tec CO6, L4

Q3) Analyze the following coding and answer the question:

If PARROT is written as 269951 and GRAND is written as 79684, how is PANT coded?

- (a) 2681 (b) 2641 (c) 2651 (d) 2645 CO6, L4

Q4) Analyze which of the following words will come third in the English dictionary?

- (a) Gerund (b) German (c) Gerbil (d) Gerent CO6, L4

Q5) Point out if the letters in the word 'UNDERSTANDING' are rearranged in the alphabetical order, which one will be in the middle in order after the rearrangement?

- (a) T (b) R (c) N (d) S CO6, L4

Q6) Explore the missing number in the series?

B6, D13, F27, H55, J111,

- (a) L313 (b) L223 (c) L181 (d) L193 CO6, L4

Q7) If the letters in the word WATER are rearranged in the alphabetical order, which one will be in the middle after the rearrangement?

- (a) A (b) W (c) R (d) None CO6, L4

Q8) Which letter in the word ALIGN occupies the same position as it does in the English alphabet?

- (a) I (b) L (c) A (d) None CO6, L4

Q9) In a certain code language FOR is written as ROF. How will PEA be written in that code language?

- (a) APE (b) AEP (c) APP (d) None CO6, L4

Q10) In a certain code 'SEQUENCE' is coded as 'FDOFVRFT'. How is 'CHILDREN' coded in that code?

- (a) OFSEMJD (b) OFSEMJE (c) OFSEMJIG (d) NONE CO6, L4

Q11) Identify loss or gain percent if on selling 15 articles, a merchant loses equal to cost price of 12 articles.

- (a) 72% gain (b) 86% gain (c) 80% loss (d) 85% loss

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CO2, L2

Q12) A certain amount earns simple interest of Rs. 2450 after 7 years. Had the interest been 3% more, how much more interest would it have earned?

- (a) 35 (b) 245 (c) 350 (d) Can not be determined

CO2, L2

Q13) A sells an item at a profit of 10% to B and B sells it to C at a profit of 30%. Find the resultant profit percent?

- (a) 23 percent (b) 43 percent (c) 33 percent (d) None

CO2, L2

Q14) A sells an item at a profit of 40% to B and B sells it to C at a loss of 10%. Find the resultant profit percent?

- (a) 26 percent (b) 33 percent (c) 81 percent (d) None

CO2, L2

Q15) Raghav spends 80% of his income. If his income increases by 12% and his expenditure increases by 17.5%, then what is the percentage decrease in his savings?

- (a) 10% (b) 20% (c) 30% (d) none of these

CO1, L1

Q16) There is 60% increase in an amount in 6 years at simple interest. What will be the compound interest of Rs. 12,000 after 3 years at the same rate

- (a) 2972 (b) 3972 (c) 4972 (d) 5972

CO1, L1

Q17) If the cost price is 80% of selling price. Then what is the profit percent?

- (a) 25 percent (b) 45 percent (c) 88 percent (d) None

CO5, L2

Q18) If a sum of money at simple interest doubles in 7 years, it will become 4 times in:

- (a) 21 years (b) 29 years (c) 9 years (d) None

CO2, L2

Q19) The difference between the C.I. and S.I. on a certain sum of money at 10% per annum for 2 years is Rs 89.92. Find the sum.

- (a) 8992 (b) 1298 (c) 2399 (d) None

CO2, L2

Q20) John borrowed Rs. 2,10,000 from a bank at an interest rate of 10% per annum, compounded annually. The loan was repaid in two equal instalments, the first after one year and the second after another year. The first instalment was interest of one year plus part of the principal amount, while the second was the rest of the principal amount plus due interest thereon. Then each instalment, in Rs., is

- (a) 121000 (b) 120000 (c) 115000 (d) 10000

CO5, L2

Q21) Determine the age of the teacher (in years) if the average age of 30 students is 12 years and when the age of their teacher is included, it becomes 13 years.

- (a) 35 (b) 39 (c) 42 (d) 43

CO1, L3

Q22) Determine the third number if the average of three numbers is 20 and the remaining two numbers are 18 and 26,

- (a) 16 (b) 18 (c) 19 (d) 20

CO1, L3

Q23) Compute the average of first 98 even numbers

- (a) 99 (b) 77 (c) 33 (d) None

CO3, L3

Q24) Determine the smallest number if average of five consecutive natural number is 93

- (a) 91 (b) 81 (c) 71 (d) None

CO3, L3

Q25) Determine the HCF of 24, 60, 48, 72

- (a) 12 (b) 18 (c) 20 (d) None

CO3, L3

Q26) How many positive integers are there from 0 to 1000 that leave a remainder of 3 on division by 7 and a remainder of 2 on division by 4

- (a) 32 (b) 36 (c) 30 (d) 18

CO3, L3

Q27) LCM of 2 natural numbers p and q where $p > q$ is 935. What is the maximum possible sum of the digits of q

- (a) 1 (b) 16 (c) 8 (d) 2

CO3, L3

Q28) Find the smallest number that leaves a remainder of 4 on division by 5, 5 on division by 6, 6 on division by 7, 7 on division by 8 and 8 on division by 9?

- (a) 2519 (b) 2509 (c) 2521 (d) 2522

CO3, L3

Q29) What are the last two digits of the number 7^{45}

- (a) 17 (b) 27 (c) 07 (d) none

CO3, L3

Q30) A 4-digit number of the form $aabb$ is a perfect square. What is the value of $a - b$?

- (a) 1 (b) 2 (c) 3 (d) 4

CO3, L3

Q31) Find the next term of the series 27, 35, 43, 51, 59,

- (a) 45 (b) 67 (c) 87 (d) none

CO6, L4

Q32) Find the next term of the series 7, 35, 14, 70, 28, 140, 56,

- (a) 280 (b) 65 (c) 22 (d) None

CO6, L4

Q33) Find the next term of the series 4, 20, 100, 500,

- (a) 2000 (b) 2500 (c) 4000 (d) None

CO6, L4

Q34) Calculate the next term of series 2500, 500, 100, 20,

- (a) 4 (b) 1 (c) 11 (d) None

CO6, L4

Q35) Calculate the next term of series 3600, 1800, 1600, 800, 600, 300,

- (a) 700 (b) 100 (c) 890 (d) None

CO6, L4

Q36) If white is called blue, blue is called red, red is called yellow, yellow is called green, green is called black, black is called violet and violet is called orange, what would be the color of grass?

- (a) Green (b) Black (c) Red (d) None

CO6, L4

Q37) If the animals which can walk are called swimmers, animals who crawl are called flying, those living in water are called snakes and those which fly in the sky are called hunters, then what will a Dog be called?

- (a) Swimmers (b) Hunters (c) Snakes (d) None

Q38) In a certain code, 2 is coded as P, 3 as N, 9 as Q, 5 as R, 4 as A and 6 as B. How is 96355 coded in that code? (a) QBNRR (b) RRNBQ (c) BNQRR (d) None

CO6, L4

Q39) In a certain code, BCA is written as CDB. How MCA is written in that code?

- (a) NDB (b) DBN (c) DER (d) none

CO6, L4

Q40) In a certain code, LPU is written as KOT. How THE is written in that code?

- (a) SGD (b) GGG (c) DGS (d) None

CO6, L4

Q41) A earns 25% more than B. C earns 25% more than A. A earns 20% more than D. E earns 20% more than A. A, B, C, D, and E earn integer amounts less than Rs. 100. What is the total amount earned by all five of them put together?

- (a) 305 (b) 310 (c) 300 (d) NONE

CO2, L2

Q42) Class B has 50% more students than class A. Number of girls in class A is equal to number of boys in class B. The percentage of girls is the same in both classes. What percentage of the student group are boys?

- (a) 40% (b) 30% (c) 35% (d) 40%

CO2, L2

Q43) A two digit number ab is 60% of x . The two-digit number formed by reversing the digits of ab is 60% more than x . Find x .

- (a) 45 (b) 54 (c) 65 (d) 56

CO2, L2

Q44) Ram sells onions in the streets of Chandni Chowk. Due to recent shortfall in the supply of onions, he doubles his selling price despite the cost price remains same for him due to a fixed price contract. He realizes that his profit have tripled. Find the original profit percent

- (a) 100% (b) 200% (c) 105% (d) 50%

CO2, L2

Q45) In a tournament, a team has played 40 matches so far and won 30% of them. If they win 60% of the remaining matches, their win percentage will be 50%. Suppose they win 90% of the remaining matches, then the total number of matches won by the team in tournament will be

- (a) 84 (b) 74 (c) 64 (d) 54

CO2, L2

Q46) Anil buys 12 toys and labels each with the same selling price. He sells 8 toys initially at 20% discount on the labeled price. Then he sells the remaining 4 toys at an additional 25% discount on the discounted price. Thus, he gets a total of Rs 2112, and makes a 10% profit. With no discounts, his percentage of profit would have been

- (a) 40 (b) 50 (c) 60 (d) 70

CO2, L2

Q47) Mukesh purchased 10 bicycles in 2017, all at the same price. He sold six of these at a profit of 25% and the remaining four at a profit of 25%. If he made a total profit of Rs. 2000, then his purchase price of a bicycle, in Rupees, was

- (a) 4000 (b) 2000 (c) 10000 (d) NONE

CO2, L2

Q48) The manufacturer of a table sells it to a wholesale dealer at a profit of 10%. The wholesale dealer sells the table to a retailer at a profit of 30%. Finally, the retailer sells it to a customer at a profit of 50%. If the customer pays Rs 4290 for the table, then its manufacturing cost (in Rs) is

- (a) 2000 (b) 3000 (c) 4000 (d) 6000

CO5, L2

Q49) If Fatima sells 60 identical toys at a 40% discount on the printed price, then she makes 20% profit. Ten of these toys are destroyed in a fire. While selling the rest, how much discount should be given on the printed price so that she can make the same amount of profit?

- (a) 28% (b) 26% (c) 22% (d) 20%

CO5, L2

Q50) In the first 10 overs of a cricket game, the run rate was only 3.2. What should be the run rate in the remaining 40 overs to reach the target of 282 runs

- (a) 6.25 (b) 6.15 (c) 6.20 (d) 6

CO5, L2

Q51) Examine which term of the series 1, 3, 5, ... will be 79?

- (a) 36th (b) 40th (c) 38th (d) 39th

CO1, L3

Q52) Determine the fourth term of a geometric progression, whose first term is 3 and the common ratio is 5.

- (a) 254 (b) 375 (c) 288 (d) 320

CO1, L3

Q53) Examine the sum of values of N for which the number 98752N is exactly divisible by 4?

- (a) 12 (b) 8 (c) 4 (d) 10

CO1, L3

Q54) Determine HCF of $(\frac{2}{4}, \frac{8}{12}, \frac{16}{20}, \frac{12}{15})$ is

- (a) 12 (b) $\frac{1}{30}$ (c) $\frac{2}{56}$ (d) $\frac{1}{60}$

CO1, L3

Q55) Compute the maximum possible length of a scale required to exactly measure 176 cm, 110 cm and 154 cm?

- (a) 10 (b) 22 (c) 11 (d) 88

CO1, L3

Q56) Compute the remainder when 2^{35} is Divided by 9?

- (a) 5 (b) 0 (c) 8 (d) 1

CO1, L3

Q57) Examine what is the highest power of 72 which can divide 70! Completely.

- (a) 16 (b) 22 (c) 67 (d) 32

CO1, L3

Q58) Compute the unit digit for $235^{(1982)} \times 784^{(2023)}$

- (a) 2 (b) 5 (c) 0 (d) 1

CO1, L3

Q59) Compute the unit digit of the product $2^{(2010)} \times 6^{(2011)} \times 7^{(2012)} \times 8^{(2013)}$

- (a) 2 (b) 4 (c) 8 (d) 0

CO1, L3

Q60) Examine if Vivek Bindra wants to count number of zeros at the end of 150!. How many are there?

- (a) 37 (b) 38 (c) 39 (d) 40

CO1, L3

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Registration No. 

Course Code: PEA305

Course Title: ANALYTICAL SKILLS-I

Paper Code A

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- Q22) Find the next term of the series 7, 35, 14, 70, 28, 140, 56,
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(a) 2000 (b) 2500 (c) 4000 (d) None CO6, L4
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(a) 700 (b) 100 (c) 890 (d) None CO6, L4
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- Q49) The difference between the C.I. and S.I. on a certain sum of money at 10% per annum for 2 years is Rs 89.92. Find the sum.
(a) 8992 (b) 1298 (c) 2399 (d) None CO2, L2

Q45) In a tournament, a team has played 40 matches so far and won 30% of them. If they win 60% of the remaining matches, then the total number of matches won by the team in the tournament will be

- (a) 84 (b) 74 (c) 64 (d) 54

CO2, L2

Q46) Anil buys 12 toys and labels each with the same selling price. He sells 8 toys initially at 20% discount on the labeled price. Then he sells the remaining 4 toys at an additional 25% discount on the discounted price. Thus, he gets a total of Rs 2112, and makes a 10% profit. With no discounts, his percentage of profit would have been

- (a) 40 (b) 50 (c) 60 (d) 70

CO2, L2

Q47) Mukesh purchased 10 bicycles in 2017, all at the same price. He sold six of these at a profit of 25% and the remaining four at a loss of 25%. If he made a total profit of Rs. 2000, then his purchase price of a bicycle, in Rupees, was

- (a) 4000 (b) 2000 (c) 10000 (d) NONE

CO2, L2

Q48) The manufacturer of a table sells it to a wholesale dealer at a profit of 10%. The wholesale dealer sells the table to a retailer at a profit of 30%. Finally, the retailer sells it to a customer at a profit of 50%. If the customer pays Rs 4290 for the table, then its manufacturing cost (in Rs) is

- (a) 2000 (b) 3000 (c) 4000 (d) 6000

CO5, L2

Q49) If Fatima sells 60 identical toys at a 40% discount on the printed price, then she makes 20% profit. Ten of these toys are destroyed in fire. While selling the rest, how much discount should be given on the printed price so that she can make the same amount of profit?

- (a) 28% (b) 26% (c) 22% (d) 20%

CO5, L2

Q50) In the first 10 overs of a cricket game, the run rate was only 3.2. What should be the run rate in the remaining 40 overs to reach the target of 282 runs

- (a) 6.25 (b) 6.15 (c) 6.20 (d) 6

CO5, L2

Q51) Examine which term of the series 1, 3, 5, will be 79?

- (a) 36th (b) 40th (c) 38th (d) 39th

CO1, L3

Q52) Determine the fourth term of a geometric progression, whose first term is 3 and the common ratio is 5.

- (a) 254 (b) 375 (c) 288 (d) 320

CO1, L3

Q53) Examine the sum of values of N for which the number 98752N is exactly divisible by 4?

- (a) 12 (b) 8 (c) 4 (d) 10

CO1, L3

Q54) Determine HCF of (2/4, 8/12, 16/20, 12/15) is

- (a) 12 (b) 1/30 (c) 2/56 (d) 1/60

CO1, L3

Q55) Compute the maximum possible length of a scale required to exactly measure 176 cm, 110 cm and 154 cm?

- (a) 10 (b) 22 (c) 11 (d) 88

CO1, L3

Q56) Compute the remainder when $2^{(35)}$ is Divided by 9?

- (a) 5 (b) 0 (c) 8 (d) 1

CO1, L3

Q57) Examine what is the highest power of 72 which can divide 70! Completely.

- (a) 16 (b) 22 (c) 67 (d) 32

CO1, L3

Q58) Compute the unit digit for $235^{(1982)} \times 784^{(2023)}$

- (a) 2 (b) 5 (c) 0 (d) 1

CO1, L3

Q59) Compute the unit digit of the product $2^{(2010)} \times 6^{(2011)} \times 7^{(2012)} \times 8^{(2013)}$

- (a) 2 (b) 4 (c) 8 (d) 0

CO1, L3

Q60) Examine if Vivek Bindra wants to count number of zeros at the end of 150!. How many are there?

- (a) 37 (b) 38 (c) 39 (d) 40

CO1, L3

Q11) Identify loss or gain percent if on selling 15 articles, a merchant loses equal to cost price of 12 articles.

- (a) 72% gain (b) 86% gain (c) 80% loss (d) 85% loss

CO2, L2

Q12) A certain amount earns simple interest of Rs. 2450 after 7 years. Had the interest been 3% more, how much more interest would it have earned?

- (a) 35 (b) 245 (c) 350 (d) Can not be determined

CO2, L2

Q13) A sells an item at a profit of 10% to B and B sells it to C at a profit of 30%. Find the resultant profit percent?

- (a) 23 percent (b) 43 percent (c) 33 percent (d) None

CO2, L2

Q14) A sells an item at a profit of 40% to B and B sells it to C at a loss of 10%. Find the resultant profit percent?

- (a) 26 percent (b) 33 percent (c) 81 percent (d) None

CO2, L2

Q15) Raghu spends 80% of his income. If his income increases by 12% and his expenditure increases by 17.5%, then what is the percentage decrease in his savings?

- (a) 10% (b) 20% (c) 30% (d) none of these

CO1, L1

Q16) There is 60% increase in an amount in 6 years at simple interest. What will be the compound interest of Rs. 12,000 after 3 years at the same rate

- (a) 2972 (b) 3972 (c) 4972 (d) 5972

CO1, L1

Q17) If the cost price is 80% of selling price. Then what is the profit percent?

- (a) 25 percent (b) 45 percent (c) 88 percent (d) None

CO5, L2

Q18) If a sum of money at simple interest doubles in 7 years, it will become 4 times in

- (a) 21 years (b) 29 years (c) 9 years (d) None

CO2, L2

Q19) The difference between the C.I. and S.I. on a certain sum of money at 10% per annum for 2 years is Rs 89.92. Find the sum.

- (a) 8992 (b) 1298 (c) 2399 (d) None

CO2, L2

Q20) John borrowed Rs. 2,10,000 from a bank at an interest rate of 10% per annum, compounded annually. The loan was repaid in two equal instalments, the first after one year and the second after another year. The first instalment was interest of one year plus part of the principal amount, while the second was the rest of the principal amount plus due interest thereon. Then each instalment, in Rs., is

- (a) 121000 (b) 120000 (c) 115000 (d) 10000

CO5, L2

Q21) Determine the age of the teacher (in years) if the average age of 30 students is 12 years and when the age of their teacher is included, it becomes 13 years.

- (a) 35 (b) 39 (c) 42 (d) 43

CO1, L3

Q22) Determine the third number if the average of three numbers is 29 and the remaining two numbers are 18 and 26.

- (a) 16 (b) 18 (c) 19 (d) 20

CO1, L3

Q23) Compute the average of first 98 even numbers

- (a) 99 (b) 77 (c) 33 (d) None

CO3, L3

Q24) Determine the smallest number if average of five consecutive natural number is 93

- (a) 91 (b) 81 (c) 71 (d) None

CO3, L3

Q25) Determine the HCF of 24, 60, 48, 72

CO3, L3

Q26) How many positive integers are there from 0 to 1000 that leave a remainder of 3 on division by 7 and a remainder of 2 on division by 4

- (a) 32 (b) 36 (c) 30 (d) 18

Q27) LCM of 2 natural numbers p and q where $p > q$ is 915. What is the maximum possible sum of the digits of q

- (a) 1 (b) 16 (c) 8 (d) 2

Q28) Find the smallest number that leaves a remainder of 4 on division by 5, 5 on division by 6, 6 on division by 7, 7 on division by 8 and 8 on division by 9?

- (a) 2519 (b) 2509 (c) 2521 (d) 2522

Q29) What are the last two digits of the number 7^{64}

- (a) 17 (b) 27 (c) 07 (d) none

Q30) A 4-digit number of the form $abab$ is a perfect square. What is the value of $a - b$?

- (a) 1 (b) 2 (c) 3 (d) 4

Q31) Find the next term of the series 27, 35, 43, 51, 59, ...

- (a) 45 (b) 67 (c) 87 (d) none

Q32) Find the next term of the series 7, 35, 14, 70, 28, 140, 56, ...

- (a) 280 (b) 65 (c) 22 (d) None

Q33) Find the next term of the series 4, 20, 100, 500, ...

- (a) 2000 (b) 2500 (c) 4000 (d) None

Q34) Calculate the next term of series 2500, 500, 100, 20, ...

- (a) 4 (b) 1 (c) 11 (d) None

Q35) Calculate the next term of series 3600, 1800, 1600, 800, 600, 300, ...

- (a) 700 (b) 100 (c) 890 (d) None

Q36) If white is called blue, blue is called red, red is called yellow, yellow is called green, green is called black, black is called violet and violet is called orange, what would be the color of grass

- (a) Green (b) Black (c) Red (d) None

Q37) If the animals which can walk are called swimmers, animals who crawl are called flying, those living in water are called snakes and those which fly in the sky are called hammers, then what will a Dog be called?

- (a) Swimmers (b) Hammers (c) Snakes (d) None

Q38) In a certain code, 2 is coded as P, 3 as N, 9 as Q, 5 as R, 4 as A and 6 as B. How is 96355 coded in that code? (a) QBNRR (b) RBNQB (c) BNQBR (d) None

Q39) In a certain code, BCA is written as CDB. How MCA is written in that code?

- (a) NDB (b) DBN (c) DER (d) none

Q40) In a certain code, LPU is written as KOT. How THE is written in that code?

- (a) SGD (b) GGG (c) DGS (d) None

Q41) A earns 25% more than B. C earns 25% more than A. A earns 20% more than D. E earns 20% more than A. A, B, C, D, and E earn integer amounts less than Rs. 100. What is the total amount earned by all five of them put together

- (a) 305 (b) 310 (c) 300 (d) None

Q42) Class B has 50% more students than class A. Number of girls in class A is equal to number of boys in class B. The percentage of girls is the same in both classes. What percentage of the student group are boys?

- (a) 40% (b) 30% (c) 35% (d) 40%

Q43) A two digit number ab is 60% if x . The two-digit number formed by reversing the digits of ab is 60% more than x . Find x .

- (a) 45 (b) 54 (c) 65 (d) 56

Q44) Ram sells onions in the streets of Chandni Chowk. Due to recent shortfall in the supply of onions, he doubles his selling price despite the cost price remains same for him due to a fixed price contract. He realizes that his profit have tripled. Find the original profit percent

- (a) 100% (b) 200% (c) 105% (d) 50%

www.devocode.in

CO3, L3

CO3, L3

CO3, L3

CO3, L3

CO3, L3

CO6, L4

CO6, L4

CO6, L4

CO6, L4

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CO6, L4

CO6, L4

CO6, L4

CO2, L2

CO2, L2

CO2, L2

CO2, L2

CO2, L2

Q56) Look at this series: 8, 6, 9, 23, 87, ... What number should come next?

A. 128 B. 226 C. 324 D. 429

(a) a (b) b (c) c (d) d

CO2, L3

Q57) If code P is denoted by 7, X by 9, M by 5, Z by 8, L by 2, T by 1, then ZLTPXM will be

(a) 812851 (b) 821591 (c) 812715 (d) 821795

CO3, L3

Q58) What letters will fill the blank?

Z, X, V, T, R, _____

(a) M,N (b) P,N (c) O,K (d) K,S

CO2, L3

Q59) Following are the criteria for selection of computer professionals in an organisation: The candidate must- 1. be a computer Engineer or MCA with first class having minimum 65% marks. 2. have secured at least 50% marks in the Selection Test. 3. have secured at least 40% marks in the interview. 4. not be less than 21 years and not more than 30 years of age as on 1-10-2005. In case of candidate who satisfies all other criteria except: A. at (1) above but is an Electronics Engineer with 70% marks, the case may be referred to the GM, Recruitment. B. At (2) above but is having at least 2 years' experience of working as a Systems Analyst, the case may be referred to the Chairman, Recruitment Committee. In each of the following questions, information about one candidate is given. You have to analyse it with reference to the above criteria and condition and then decide the appropriate course of action. You are not to assume anything other than the given information. All these cases are given to you as on 1.10.2005.
Rohit verma is a Mechanical Engineer with 75% marks. He was born on 6th July, 1976. He scored 66% marks in the Selection Test and 52% marks in the interview.

(a) To be selected (b) Not to be selected (c) To be Referred to the Chairman (d) To be referred to GM

CO2, L3

Q60) Find the next number in the series 5, 8, 16, 19, 57, 60,

(a) 188 (b) 256 (c) 240 (d) 180

CO2, L3

—End of Question paper—

Q43) There is 60% increase in an amount in 6 years at simple interest. What will be the compound interest of Rs. 12,000 after 3 years at the same rate

- (a) 2972 (b) 3972 (c) 4972 (d) 5972

CO1, L1

Q44) A man sold two houses for Rs. 8.6 lakhs each. On the one, he gained 10% and on the other, he lost 10%. What percent is the effect of the sale on the whole?

- (a) 05% Loss (b) 0.5% Profit (c) 1% Loss (d) No Profit No Loss

CO1, L1

Q45) A Shopkeeper buys two bicycles for Rs. 1500. He sells first bicycle at a profit of 22% and the second bicycle at a loss of 8%. What is the SP of second bicycle if in the whole transaction there is no profit no loss?

- (a) 1012 (b) 1100 (c) 1245 (d) 1250

CO1, L1

Q46) Sam purchased 20 dozens of toys at the rate of Rs. 375 per dozen. He sold each one of them at the rate of Rs. 33. What was his percentage profit?

- A. 3.5 B. 4.5 C. 5.6 D. 6.5

- (a) a (b) b (c) c (d) d

CO1, L1

Q47) A boy sold a pen at a profit of 10% if the cost price of the pen is Rs. 120. What is selling price of the pen?

- (a) Rs. 213 (b) Rs. 132 (c) Rs. 110 (d) Rs. 135

CO1, L1

Q48) A sum was lent at 20% p.a compound interest, the interest being compounded annually. Rs. 2400 was paid back after 1 year. After another year Rs. 2880 was repaid to clear the loan. Find the sum lent

- (a) 2000 (b) 3000 (c) 4000 (d) None of these

CO1, L1

Q49) The difference between the compound interest and simple interest on a certain sum at 10% per annum for 2 years is Rs. 631. Find the sum.

- (a) 63100 (b) 62100 (c) 61100 (d) 60100

CO1, L1

Q50) If a student score 15% marks and he is failed with 180 marks. But if he score 85% marks then he is pass with 240 more marks. Then find the passing percentage?

- (a) 39% (b) 45% (c) 42% (d) 48%

CO1, L1

Q51) Find the missing number in the following series?

3, 5, 5, 19, 7, 41, 9, ?

- (a) 71 (b) 67 (c) 53 (d) 72

CO2, L3

Q52) Which word will come in the middle if all of them are arranged alphabetically as in a dictionary?

- (a) Praise (b) Practical (c) Prank (d) Prayer

CO2, L3

Q 53. Analyse the information given below about an election campaign strategy and answer the following questions: You are supposed to prepare a strategy for election campaigning which is going to start in 6 months time. This is to be done at multiple constituencies involving many agencies. There would be a huge amount of logistical support which would be required once the format has been decided. The Minister has called for a meeting to hear your action plan and you need to be prepared for it.

Q. What would be the most effective answer?

- (a) Concentrate on coordinating with the different agencies and having them approve the strategy
(b) Concentrate on setting deadlines, targets and checkpoints in order to meet the deadlines.
(c) Concentrate on getting an immediate feel for the contents design and layout of the strategy
(d) Concentrate on logistics: how to move people and equipments from one place to another smoothly

CO2, L3

Q54. SAB, ?, QCD, PDD, OEF, NFF

- (a) GHB (b) RBB (c) RBC (d) RUB

CO2, L3

Q55) QAR, RAS, SAT, TAU, _____

- A. UAV B. UAT C. TAS D. TAT

- (a) a (b) b (c) c (d) d

CO2, L3

A student scored 30% in an examination and still failed by 12 marks while another candidate scored 40% and got 28 marks more than the passing marks. Then find the maximum marks in the examination.

CO1, L1

- (a) 55 (b) 77 (c) 400 (d) 888

The current population of a town is 66,000. During the last 2 years the population increased at the rate of 10% and 20% per year. The population 2 years ago was (approximately)

CO1, L1

- (a) 50000 (b) 70000 (c) 68999 (d) None

In an election of 2 candidate 12% of the voters did not cast their votes. The winner won by getting 45% of the total votes, defeated his rival by 2000 votes. Find the total number of voters.

CO1, L1

- (a) 100000 (b) 440000 (c) 450000 (d) None

CP of 25 articles is equal to SP of 20 articles. Find the profit or loss percent.

CO1, L1

- (a) 25% Profit (b) 35% Profit (c) 25% Loss (d) None

A shopkeeper buys some number of article at the rate of 11 articles for Rs.10 and sold all of them at the rate of 10 articles for Rs.11. Find his profit or loss percent.

CO1, L1

- (a) 21% Profit (b) 22% Profit (c) 21% Loss (d) None

A shopkeeper purchases some number of article for Rs.8400. He sold $\frac{3}{5}$ th of them at 15% profit each. At what percent profit should he sell the remaining to gain overall 20% profit.

CO1, L1

- (a) 27.5% (b) 30% (c) 40% (d) None

A shopkeeper professes to sell his goods at 10% profit, But he uses 20% less weight. Find his total profit percent.

CO1, L1

- (a) 35% (b) 37.5% (c) 23% (d) None

Buy 5 articles and get 3 articles free. Find discount percent.

CO1, L1

- (a) 33% (b) 37.5% (c) 43 (d) None

If RETURN is coded as PGRWPP, how is SANDLE coded as?

CO2, L3

- (a) UXPBNC (b) QCLFJG (c) CNBPXU (d) CLGFQB

If BRAND is written as 79643 and PAROT is written as 26951, how is PANT coded?

CO2, L3

- (a) 2641 (b) 2643 (c) 3451 (d) 7423

If nit zec bec means eat fresh food,

get pec jac means food is good,

tac lac mac means she is nice,

then which word means good?

- (a) zec (b) bec (c) nit (d) pec

CO2, L3

In a code language if R= 9 and DATE= 15, then what will be the code of ACT.

CO2, L3

- (a) 12 (b) 13 (c) 14 (d) 8

In a certain code, '154' means 'Ram is good'; '537' means 'shyam is bud' and '472' means 'good and bad'. Which of the following represents 'shyam' in that code?

CO2, L3

- (a) 5 (b) 1 (c) 3 (d) 2

If AC is called TV, TV is called fridge, fridge is called washing machine, washing machine is called computer, computer is called phone, phone is called printer and printer is called calculator then what would be used to wash clothes?

CO2, L3

- (a) TV (b) Calculator (c) Computer (d) Printer

In a certain code 'LAKE' is written as \$ #1& and TESLA is written as @&%\$# How can 'TASK' be written in that code?

CO2, L3

- (a) #@\$! (b) @%#&\$
(c) @#%! (d) none

Q 28 In the series given below, Find the missing no?

25, 49, ..., 169, 289

- (a) 100 (b) 111 (c) 144 (d) 121

CO2, L3

Q 29 In the series given below,

Find the missing no?

BAC, ..., NMO, TSU, ZYA

- (a) HIG (b) IHG (c) HGI (d) IGH

CO2, L3

Q 30) 30, 68, 130, 222, ?, 520, 738.

- (a) 420 (b) 350 (c) 250 (d) 280

CO2, L3

Q31) Find the least value of * for which $1*74729$ is divisible by 9.

- (a) 1 (b) 2 (c) 3 (d) 4

CO4, L2

Q32) There are 36 students in a class A whose average weight is 30kg and 24 students in class B whose average is 40kg. What will be the average if the classes are combined?

- (a) 34 (b) 45 (c) 52 (d) None

CO4, L2

Q33) $N = 25677AB$, N is divisible by 36 what is the sum of the digits of N ?

- (a) 36 (b) 35 (c) 45 (d) N.O.T

CO4, L2

Q34) Find the highest power of 12 in $60!$?

- (a) 23 (b) 28 (c) 25 (d) N.O.T

CO4, L2

Q35) The average weight of 4 men is increased by 8 kg, when one of them whose weight is 75 kg is replaced by another women. What is the weight of the new women (in kg)?

a. 100 b. 107 c. 120 d. 101

- (a) a (b) b (c) c (d) d

CO4, L2

Q36) Find the unit digit of $376!$

- (a) 0 (b) 6 (c) 2 (d) 4

CO4, L2

Q37) The greatest four digit number which is exactly divisible by each one of the numbers 12, 18, 21 and 28 is

- (a) 9828 (b) 9288 (c) 9882 (d) 9929

CO4, L2

Q38) Find the fourth term of a geometric progression, whose first term is 3 and the common ratio is 5.

- (a) 6 (b) 8 (c) 7 (d) 10

CO4, L2

Q39) Find the unit digit of 2332^{23131}

- (a) 2 (b) 4 (c) 8 (d) 6

CO4, L2

Q40) Find the first term of an AP whose 8th term is 39 and 12th term is 59.

- (a) 3 (b) 4 (c) 5 (d) 6

CO4, L2

Q41) If an article is sold at 5% gain instead of 5% loss, the seller gets Rs. 50 more. The C.P. of the article is:

- (a) 250 (b) 500 (c) 5000 (d) none of these

CO1, L1

Q42) Raghav spends 80% of his income. If his income increases by 12% and his expenditure increases by 17.5%, then what is the percentage decrease in his savings?

- (a) 10% (b) 20% (c) 30% (d) none of these

CO1, L1

COURSE CODE: PEA307

COURSE TITLE: ADVANCED ANALYTICAL SKILLS-I

Max. Marks: 60

Time Allowed: 01:30 hrs.

Read the following instructions carefully before attempting the question paper.

1. Match the Paper Code shaded on the OMR Sheet with the Paper code mentioned on the question paper and ensure that both are the same.
2. This question paper contains 60 questions of 1 mark each. 0.25 marks will be deducted for each wrong answer.
3. All questions are compulsory.
4. Do not write or mark anything on the question paper and/or on rough sheet(s) which could be helpful to any student in copying, except your registration number on the designated space.
5. Submit the question paper and the rough sheet(s) along with the OMR sheet to the invigilator before leaving the examination hall.
6. Use of calculators is prohibited.

Q 1 What will be the remainder if $55555 \dots 20$ times is divided by 13?

CO4, L2

- (a) 2 (b) 3 (c) 7 (d) None of these

Q 2 What will be the remainder if 23^{49} is divided by 13?

CO4, L2

- (a) 8 (b) 9 (c) 10 (d) 11

Q 3 Find the unit digit of 12^{45} ?

CO4, L2

- (a) 6 (b) 2 (c) 8 (d) 4

Q 4 Find the number of zeroes at the trail of $120!$?

CO4, L2

- (a) 24 (b) 26 (c) 27 (d) 28

Q 5 Find the highest power of 6 in $50!$?

CO4, L2

- (a) 15 (b) 18 (c) 22 (d) None of these

Q 6 If $848A6A$ is divisible by 11, then value of A is:

CO4, L2

- (a) 8 (b) 7 (c) 9 (d) 2

Q 7 How many odd factors of $7^4 \times 3^2 \times 2^8$ are there?

CO4, L2

- (a) 10 (b) 15 (c) 135 (d) 64

Q 8 The sum of third and ninth term of an A.P is 8. Find the sum of the first 11 terms of the progression.

CO4, L2

- (a) 11 (b) 22 (c) 33 (d) 44

Q 9 What is the sum of the following series? $-64, -66, -68, \dots, -100$

CO4, L2

- (a) -1458 (b) -1558 (c) -1568 (d) -1664

Q 10 Find the sum of the geometric series:

CO4, L2

- $4 - 12 + 36 - 108 + \dots$ upto 10 terms
- (a) -59048 (b) -59343 (c) -51047 (d) -59088

Q 11 If A's salary is 20 % more than B, then by how much Percent B's salary is less than A.

CO1, L1

- (a) 16.66% (b) 18% (c) 22% (d) 39%

Q 12 If the price of sugar is reduced by 20% due to which a person can buy 2kg more sugar for Rs. 200. Find the original price of sugar per kg.

CO1, L1

- (a) 25 (b) 55 (c) 76 (d) None

Time Allowed: 01:30 hrs.

Read the following instructions carefully before attempting the question paper

1. Match the Paper Code stated on the OMR Sheet with the Paper code mentioned on the question paper and ensure that both are the same.
2. This question paper contains 30 questions of 1 mark each. 0.25 marks will be deducted for each wrong answer.
3. All questions are compulsory.
4. Do not write or mark anything on the question paper and/or on rough sheets which could be helpful to any student in copying, except your registration number on the designated space.
5. Submit the question paper and the rough sheet(s) along with the OMR sheet to the invigilator before leaving the examination hall.
6. Use of calculator is prohibited.

Q1) Analyse the following coding:

In a code language, PROPER is coded as NDMRT, how is RETURN coded in the same code?

- (a) TCVSTL (b) PGRWQB (c) CNBPNU (d) PGRWQB

CO6, L4

Q2) Analyse the following coding and answer the question

Ert ruc lnc means our fresh food.

ruc lnc jnc means food is tasty.

jnc lnc ruc means the is beautiful.

Then which word means tasty?

- (a) jnc (b) lnc (c) ruc (d) erc

CO6, L4

Q3) Analyse the following coding and answer the question

IFPABRUT is written as 259951 and GRAMES is written as 79684, how is PANT coded?

- (a) 2681 (b) 2641 (c) 2621 (d) 2643

CO6, L4

Q4) Analyse which of the following words will come third in the English dictionary?

- (a) Germsl (b) German (c) Germl (d) Gerent

CO6, L4

Q5) If the letters in the word 'UNDERSTANDING' are rearranged in the alphabetical order, which one will be in the middle in order after the rearrangement?

- (a) T (b) R (c) N (d) S

CO6, L4

Q6) Explore the missing number in the series?

36, 121, 177, 122, 111, ...

- (a) 513 (b) 122 (c) 1181 (d) 1193

CO6, L4

Q7) If the letters in the word 'WATER' are rearranged in the alphabetical order, which one will be in the middle after the rearrangement?

- (a) A (b) W (c) R (d) None

CO6, L4

Q8) Which letter in the word ALIGN occupies the same position as it does in the English alphabet?

- (a) I (b) L (c) A (d) None

CO6, L4

Q9) In a certain code language FOR is written as ROF. How will PEA be written in that code language?

- (a) APE (b) AEP (c) APP (d) None

CO6, L4

Q10) In a certain code 'SEQUENCE' is coded as FDOFVRFT. How is 'CHILDREN' coded in that code?

- (a) OFSENUIT (b) OFSENUITE (c) OFSENUIG (d) NONE

CO6, L4

Q43) In a tournament, a team has played 40 matches so far and won 30% of them. If they win 60% of the remaining matches, their win percentage will be 50%. Suppose they win 90% of the remaining matches, then the total number of matches in the tournament will be

- (a) 84 (b) 74 (c) 64 (d) 54

Q46) Anil buys 12 toys and labels each with the same selling price. He sells 8 toys initially at 20% discount on the labeled price. Then he sells the remaining 4 toys at an additional 25% discount on the discounted price. Thus, he gets a total of Rs 2112, and makes a 10% profit. With no discounts, his percentage of profit would have been

- (a) 40 (b) 50 (c) 60 (d) 70

Q47) Mukesh purchased 10 bicycles in 2017, all at the same price. He sold six of these at a profit of 25% and the remaining four at a 25% loss. If he made a total profit of Rs. 2000, then his purchase price of a bicycle, in Rupees, was

- (a) 4000 (b) 2000 (c) 10000 (d) NONE

Q48) The manufacturer of a table sells it to a wholesale dealer at a profit of 10%. The wholesale dealer sells the table to a retailer at a profit of 30%. Finally, the retailer sells it to a customer at a profit of 50%. If the customer pays Rs 4290 for the table, then its manufacturing cost (in Rs) is

- (a) 2000 (b) 3000 (c) 4000 (d) 6000

Q49) If Fatima sells 60 identical toys at a 40% discount on the printed price, then she makes 20% profit. Ten of these toys are destroyed. While selling the rest, how much discount should be given on the printed price so that she can make the same amount of profit?

- (a) 28% (b) 26% (c) 22% (d) 20%

Q50) In the first 10 overs of a cricket game, the run rate was only 3.2. What should be the run rate in the remaining 40 overs to reach a target of 282 runs

- (a) 6.25 (b) 6.15 (c) 6.20 (d) 6

Q51) Examine which term of the series 1, 3, 5, ... will be 79?

- (a) 36th (b) 40th (c) 38th (d) 39th

Q52) Determine the fourth term of a geometric progression, whose first term is 3 and the common ratio is 5.

- (a) 254 (b) 375 (c) 288 (d) 320

Q53) Examine the sum of values of N for which the number 98752N is exactly divisible by 4?

- (a) 12 (b) 8 (c) 4 (d) 10

Q54) Determine HCF of $(\frac{2}{4}, \frac{8}{12}, \frac{16}{20}, \frac{12}{15})$ is

- (a) 12 (b) $\frac{1}{30}$ (c) $\frac{2}{56}$ (d) $\frac{1}{60}$

Q55) Compute the maximum possible length of a scale required to exactly measure 176 cm, 110 cm and 154 cm?

- (a) 10 (b) 22 (c) 11 (d) 88

Q56) Compute the remainder when 2^{35} is Divided by 9?

- (a) 5 (b) 0 (c) 8 (d) 1

Q57) Examine what is the highest power of 72 which can divide 70! Completely.

- (a) 16 (b) 22 (c) 67 (d) 32

Q58) Compute the unit digit for $235^{(1982)} \times 784^{(2023)}$

- (a) 2 (b) 5 (c) 0 (d) 1

Q59) Compute the unit digit of the product $2^{(2010)} \times 6^{(2011)} \times 7^{(2012)} \times 8^{(2013)}$

- (a) 2 (b) 4 (c) 8 (d) 0

Q60) Examine if Vivek Bindra wants to count number of zeros at the end of 150!. How many are there?

- (a) 37 (b) 38 (c) 39 (d) 40

- Q28) How many positive integers are there from 1 to 1000 that leave a remainder of 3 on division by 7 and a remainder of 2 on division by 4?
- (a) 32 (b) 36 (c) 30 (d) 18
- Q29) The LCM of 2 natural numbers p and q where $p > q$ is 935. What is the maximum possible sum of the digits of q and the smallest number that leaves a remainder of 4 on division by 5, 5 on division by 6, 6 on division by 7, 7 on division by 8 and 8 on division by 9?
- (a) 18 (b) 2509 (c) 2521 (d) 2522
- Q30) What are the last two digits of the number 7^{745} ?
- (a) 17 (b) 27 (c) 07 (d) none
- Q31) A 4-digit number of the form $aabb$ is a perfect square. What is the value of $a + b$?
- (a) 1 (b) 2 (c) 3 (d) 4
- Q32) Find the next term of the series 27, 35, 43, 51, 59, _____
- (a) 45 (b) 67 (c) 87 (d) none
- Q33) Find the next term of the series 7, 35, 14, 70, 28, 140, 56, _____
- (a) 280 (b) 65 (c) 22 (d) None
- Q34) Find the next term of the series 4, 20, 100, 500, _____
- (a) 2000 (b) 2500 (c) 4000 (d) None
- Q35) Calculate the next term of series 2500, 500, 100, 20, _____
- (a) 4 (b) 1 (c) 11 (d) None
- Q36) Calculate the next term of series 3600, 1800, 1600, 800, 600, 300, _____
- (a) 700 (b) 100 (c) 890 (d) None
- Q37) If white is called blue, blue is called red, red is called yellow, yellow is called green, green is called black, black is called violet, violet is called orange, what would be the color of grass?
- (a) Green (b) Black (c) Red (d) None
- Q38) If the animals which can walk are called swimmers, animals who crawl are called flying, those living in water are called those which fly in the sky are called hunters, then what will a Dog be called?
- (a) Swimmers (b) Hunters (c) Snakes (d) None
- Q39) In a certain code, 2 is coded as P, 3 as N, 9 as Q, 5 as R, 4 as A and 6 as B. How is 96355 coded in that code?
- (a) QBNRR (b) RRNQB (c) BNQRR (d) None
- Q40) In a certain code, BCA is written as CDB. How MCA is written in that code?
- (a) NDB (b) DBN (c) DER (d) none
- Q41) In a certain code, LPU is written as KOT. How THE is written in that code?
- (a) SGD (b) GGG (c) DGS (d) None
- Q42) A earns 25% more than B. C earns 25% more than A. A earns 20% more than D. E earns 20% more than A. A, B, C, D, E get integer amounts less than Rs. 100. What is the total amount earned by all five of them put together?
- (a) 305 (b) 310 (c) 300 (d) NONE
- Q43) Class B has 50% more students than class A. Number of girls in class A is equal to number of boys in class B. The percentage of the student group are boys?
- (a) 40% (b) 30% (c) 35% (d) 40%
- Q44) A two digit number ab is 60% of x . The two-digit number formed by reversing the digits of ab is 80% more than x . Find the value of x .
- (a) 45 (b) 54 (c) 65 (d) 56
- Q45) Ram sells onions in the streets of Chandni Chowk. Due to recent shortfall in the supply of onions, he doubles his selling price but the price remains same for him due to a fixed price contract. He realizes that his profit have tripled. Find the original profit percentage.
- (a) 100% (b) 200% (c) 105% (d) 50%

Q11) Identify loss or gain percent if on selling 15 articles, a merchant loses equal to cost price of 12 articles.

- (a) 72% gain (b) 86% gain (c) 80% loss (d) 85% loss

CO2, L3

Q12) A certain amount earns simple interest of Rs. 2450 after 7 years. Had the interest been 3% more, how much more interest would have earned?

- (a) 35 (b) 245 (c) 350 (d) Can not be determined

CO2, L3

Q13) A sells an item at a profit of 10% to B and B sells it to C at a profit of 30%. Find the resultant profit percent?

- (a) 23 percent (b) 43 percent (c) 33 percent (d) None

CO2, L3

Q14) A sells an item at a profit of 40% to B and B sells it to C at a loss of 10%. Find the resultant profit percent?

- (a) 26 percent (b) 33 percent (c) 81 percent (d) None

CO2, L3

Q15) Raghav spends 80% of his income. If his income increases by 12% and his expenditure increases by 17.5%, then what is the percentage decrease in his savings?

- (a) 10% (b) 20% (c) 30% (d) none of these

CO1, L1

Q16) There is 60% increase in an amount in 6 years at simple interest. What will be the compound interest of Rs. 12,000 after 3 years at the same rate?

- (a) 2972 (b) 3972 (c) 4972 (d) 5972

CO1, L1

Q17) If the cost price is 80% of selling price. Then what is the profit percent?

- (a) 25 percent (b) 45 percent (c) 88 percent (d) None

CO5, L3

Q18) If a sum of money at simple interest doubles in 7 years, it will become 4 times in:

- (a) 21 years (b) 29 years (c) 9 years (d) None

CO2, L3

Q19) The difference between the C.I. and S.I. on a certain sum of money at 10% per annum for 2 years is Rs 89.92. Find the sum.

- (a) 8992 (b) 1298 (c) 2399 (d) None

CO2, L3

Q20) John borrowed Rs. 2,10,000 from a bank at an interest rate of 10% per annum, compounded annually. The loan was repaid by equal instalments, the first after one year and the second after another year. The first instalment was interest of one year plus part of the principal amount, while the second was the rest of the principal amount plus due interest thereon. Then each instalment, in Rs.,

- (a) 121000 (b) 120000 (c) 115000 (d) 10000

CO5, L3

Q21) Determine the age of the teacher (in years) if the average age of 30 students is 12 years and when the age of their teacher is included, it becomes 13 years.

- (a) 35 (b) 39 (c) 42 (d) 43

CO1, L3

Q22) Determine the third number if the average of three numbers is 20 and the remaining two numbers are 18 and 28.

- (a) 16 (b) 18 (c) 19 (d) 20

CO1, L3

Q23) Compute the average of first 98 even numbers.

- (a) 99 (b) 77 (c) 33 (d) None

CO3, L3

Q24) Determine the smallest number if average of five consecutive natural number is 93.

- (a) 91 (b) 81 (c) 71 (d) None

CO3, L3

Q25) Determine the HCF of 24, 60, 48, 72.

- (a) 12 (b) 18 (c) 20 (d) None

CO3, L3